# What makes the whole? Probing Attribute-Level Compositionality in LLM Judges

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## Abstract

LLM judges produce a multidimensional evaluation in the form of an overall quality score, along with fine-grained attribute-level judgments, implicitly hinting at compositionality. We investigate whether LLM judges genuinely compose part-evaluations into whole, or instead produce decomposable-looking evaluations through a different mechanism. Across diverse multidimensional evaluation settings (summarization, dialogue, and essay), we apply a four-stage diagnostic framework of increasing strictness: (1) how well the judge agrees with humans on overall and trait scores; (2) whether the overall score can be reconstructed from the trait scores; (3) whether the inferred composition is stable and human-aligned; and (4) whether it is causally responsive to targeted trait degradation. We further identify the rubric's geometry as the factor governing all four. Across summarization, dialogue, and essay evaluation (4 models × 3 prompting strategies across 3 tasks and 4 datasets), we find that LLM judges produce highly decomposable outputs ($R^2$ = 0.755-0.934), yet their inferred composition diverges sharply from humans (Kendall's $\tau$ between LLM and human trait orderings: -0.07 to 0.20). Counterfactual perturbations for targeted trait degradation establish that decomposability does not imply causal sensitivity. The dominant factor is not model capacity, but the geometry of the rubric. The externally-imposed compositional structure explains 17× more variance in compositional behavior than model choice ($\eta^2$= 0.384 vs. 0.022). The decomposability of structured output is a weak indicator of genuine compositional reasoning, and the inductive biases imposed by the task structure can dominate the model-side capacity.

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*Accepted to the $2^{nd}$ Workshop on Compositional Learning*, at ICML 2026, Seoul, South Korea. 

## 1. Introduction

Compositionality refers to the ability to form and reason about complex concepts by combining simpler, reusable subcomponents. Although the latest LLMs and AI systems exhibit increasingly sophisticated behavior, their ability to capture and exploit compositional structure in complex real-world domains remains uncertain. We study an example of such a complex use case, viz., natural language evaluation with LLM judges. Specifically, we investigate whether, when an LLM judge is prompted to score a piece of textual content on multiple criteria and then assign an overall evaluation, it composes the overall judgment from the criterion scores in the way a human evaluator might compose?

This question is particularly pressing for multidimensional LLM evaluation, where judges produce both an overall score and fine-grained trait scores, offering more interpretable and actionable feedback than an overall score alone. However, strong overall alignment between LLM judges and human evaluators does not guarantee that attribute[1]-level evaluations are equally aligned or informative. An LLM judge with high agreement (e.g. $\rho$ = 0.70) with humans on overall quality might demonstrate divergent behavior on fine-grained attributes, or how it weighs these traits while forming the overall judgment. Current benchmarking works rarely examine this compositionality, the relationship between trait-level scores and overall assessment, and whether this relationship matches with how human evaluators view the evaluation. As multidimensional LLM evaluations become mainstream, understanding how judges arrive at a particular judgment is equally critical along with whether their agreement with human experts. We investigate whether this composition can be identified, whether it is human-aligned, and whether trait-level scores carry genuine diagnostic value beyond the overall score. Across tasks and datasets, we find that the geometry of the evaluation rubric is a primary determinant of whether multidimensional LLM judgments are composable, human-aligned, and diagnostically useful. Concretely, we ask four questions of increasing depth: how well the judge agrees with humans on overall and trait scores; whether the overall score can be rebuilt from those trait scores; whether

[1]We use traits, criteria, and attributes interchangeably in the context of evaluation dimensions.

that reconstruction is stable, matches how humans combine traits, and survives targeted degradation of a single trait; and what rubric properties govern when this composition is interpretable and diagnostically useful (summarized in Figure 1)

**RQ1**) Human-LLM alignment: How well do LLM judges align with humans on overall scores and on fine-grained trait-level scores across multidimensional evaluation settings?

**RQ2**) Decomposability Test: To what extent can an LLM judge's overall score be composed from its trait-level scores, and how does this vary across tasks and rubrics?

**RQ3**) Reality of Composition: When the composition is interpretable, is the composition stable across attribution methods, and does it align with the corresponding human evaluative structure?

**RQ4**) What governs the composition: How do rubric properties affect compositionality and interpretability, and when do trait-level scores provide information beyond the overall score?

Our contributions are as follows:

1. A compositionality diagnostic framework for LLM judges that examines the relationship between trait-level and overall scores. In contrast to previous work, we systematically analyze the compositional structure underneath the alignment.

2. We empirically demonstrate that rubric design is the primary factor of compositional behavior. Rubric structure explains 17× more variance in compositional patterns than model choice ($\eta^2 = 0.384$ vs. 0.022).

3. Counterfactual validation of trait-level sensitivity: We apply targeted perturbations for 5 essay traits and measure whether LLM judges respond specifically to the degraded trait.

## 2. Related Work

### 2.1. LLM-as-a-Judge

LLM-as-a-Judge frameworks have emerged as a scalable alternative to human evaluation across a range of generation tasks explored in (Chiang & Lee, 2023), (Yuan et al., 2023), and (Zheng et al., 2023), and studies such as (Gu et al., 2024) and (Liu et al., 2023) report strong overall agreement with human evaluators. However, most of this literature assesses judges primarily through aggregate metrics such as overall correlation or accuracy. Our work extends this line by asking not only whether judges agree with humans overall, but also how their overall judgments are composed from fine-grained evaluation traits.

### 2.2. Multidimensional Evaluation

Recent multidimensional evaluation frameworks decompose holistic quality into finer-grained criteria to improve interpretability and evaluation quality (Liu et al., 2023; Feng et al., 2025). Works such as FLASK (Fine-Grained Language Model Evaluation)from (Ye et al., 2024), LMUnit from (Saad-Falcon et al., 2024), HD-EVAL from (Liu et al., 2024), and work in (Yu et al., 2025) demonstrate the effectiveness of decomposed evaluation, enabling interpretable assessment and improved human-LLM evaluation agreement. However, these works primarily use decomposition as a design choice to improve the evaluation. In contrast, our goal is diagnostic: we study whether overall scores are recoverable from trait scores, whether the inferred composition is stable and human-aligned across attribution methods, and whether the resulting trait-level outputs provide information beyond the overall score

### 2.3. Attribution and Interpretability

Attribution methods such as SHAP in (Lundberg & Lee, 2017), LIME in (Ribeiro et al., 2016), and related interpretability tools such as token-SHAP in (Horovicz & Goldshmidt, 2024) and LAMP in (Chen et al., 2025) are widely used to identify which inputs or features drive model predictions. Our setting differs in both the unit of analysis and the purpose. Rather than attributing predictions to tokens or internal activations, we apply attribution at the level of evaluation traits, using linear decomposition, ablation, and SHAP to analyze how LLM judges form overall quality judgments from fine-grained rubric dimensions. This allows us to study not only interpretability, but also human alignment, rubric geometry, and the diagnostic value of multidimensional evaluation.

### 2.4. Compositional Generalization

Recent works investigate the compositional abilities of neural models. Five aspects of compositionality (systematicity, productivity, substitutivity, localism, overgeneralization) are formalized in (Hupkes et al., 2020), showing that standard architectures satisfy them only partially. Lake & Baroni (2018) introduce SCAN as a controlled probe of systematic generalization. The following works in (Press et al., 2023; Dziri et al., 2023) establish that even frontier LLMs demonstrate a compositionality gap where high accuracy on individual sub-tasks does not automatically imply correct composition into a multi-step solution. Our work studies the analogous question in a different setting. Instead of asking whether LLMs can compose multi-step reasoning, we investigate whether they compose evaluative judgments from rubric-defined parts. Our observations closely align with the broader finding that composite outputs need not reflect compositional mechanisms.

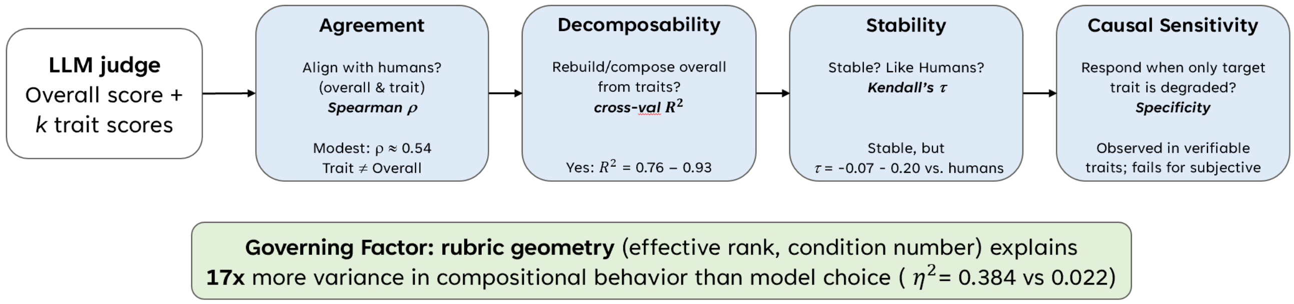


*Figure 1.* Each stage probes a stricter property — agreement, decomposability, stability and human alignment, and causal sensitivity. Rubric geometry is the dominant factor governing all four.

## 3. Methodology

**Formulation** We consider the NLG evaluation as a multi-dimensional assessment problem. Given an input and output text, an evaluator assigns an overall quality score and optionally, scores on $k$ fine-grained quality attributes. We investigate the implicit mapping between fine-grained attributes and overall score as $f : (a_1, a_2, ..., a_k) \rightarrow y_{overall}$ and whether this mapping differs between human and LLM Judges. The detailed formulation is in the Appendix.

**Alignment Measurement** We measure the agreement between LLM judges and humans at two levels of granularity, overall quality and per-trait. Since the score scales differ across datasets and several settings are ordinal, we use Spearman's rank correlation $\rho$ throughout our analysis.

**Compositionality** For each judge source $s \in \{human, llm\}$, we fit $y_{overall} = \beta_0 + \sum_{i=1}^{k} \beta_i \cdot a_i + \epsilon$, where the attribute scores are standardized to allow direct comparison of the coefficients and report cross-validated $R^2$ as the primary measure of compositionality. High $R^2$ indicates that the variation in the overall score is well explained by the corresponding attribute scores. Importantly, this measures interpretability or recoverability, not human alignment. We evaluate both linear models (linear regression, leave-one-trait out ablation) and a non-linear alternative (SHAP with Random Forest regressors) to test whether the observed part–whole structure is largely additive or depends on higher-order interactions.

**Attribution** We extract trait importance using the three attribution methods mentioned above and measure convergence using Kendall's $\tau$ for each pair. High $\tau$ indicates stable, method-independent importance estimates. Additionally, we compare the attribute-importance ranking inferred from LLM scores ($\beta_{i^l}$) with the corresponding ranking inferred from human scores ($\beta_{i^h}$). This tests whether LLMs form overall judgments in a manner similar to humans (RQ3). Together, these analyses separate decomposability from human-aligned composition.

***Alignment Diagnostic Test (ADT)***: Human-derived weights ($\beta_{i^h}$) from regression are applied to LLM trait scores to compute the adjusted overall score $y_{overall^{adjusted}} = \sum_i \beta_{i^h} \cdot a_{i^l}$ and the reweighting benefit $\Delta\rho = \rho(y_{overall^h}, y_{overall^{adjusted}}) - \rho(y_{overall^l}, y_{overall^{adjusted}})$. $\Delta\rho > 0.05$ indicates compositional weight mismatch (trait scores are reasonable but combined differently).

***Causal Compositional Probe via Counterfactual Perturbation*** The decomposability ($R^2$) and attribution-stability ($\tau$) are correlational. They do not indicate causal understanding and sensitivity. We construct a targeted-perturbation experiment as the causal complement. For each of the 5 traits in the ELLIPSE dataset, we degrade the target trait while preserving others using rule-based instructions and rescore with Claude and LLaMA. We measure (i) Target $\Delta$, i.e., the score change on the targeted trait, (ii) Non-target $\Delta$, i.e., the mean change on non-targeted traits, and (iii) Specificity as (Target $\Delta$ - Non-target $\Delta$), i.e., the marginal causal effect above the general-drift baseline. This tests whether an attribute score responds specifically to its intended dimension or whether its behavior is diffuse across the rubric.The vocabulary, with $< 0.1\%$ character change across 95% of the essays, serves as a null-control setup in which a well-calibrated judge should produce near-zero responses.

**Rubric Characterization** We categorize traits as verifiable (concrete textual referents, e.g., coverage, accuracy), subjective (holistic, no external referent, e.g., coherence, naturalness), or semi-objective (e.g., conventions, organization). Full categorization in Appendix D.

***Rubric Geometry*** Additionally, we hypothesize that multidimensional evaluation depends strongly on the rubric's geometry, i.e., the degree of independence or entanglement among the attributes (RQ4). We characterize the rubrics by effective rank (exponential entropy of normalized singular values from the trait correlation matrix — measures trait independence) and condition number ( $\sigma_{max}/\sigma_{min}$) — measures weight estimation stability). Details of computation are in the Appendix.

***Cross-rubric control*** A key question is whether the structural properties observed (e.g., ASAP++'s high condition number) are intrinsic to the task domain or to the rubric design. To separate rubric effects from content effects, we reevaluate 100 ELLIPSE essays under the ASAP++ rubric (5 traits) and compare the resulting structural properties of traits with ASAP++ on its own data. If the condition numbers match, it confirms that the structural property follows the rubric rather than the content.

***Variance decomposition*** To quantify whether compositional behavior is driven by task/rubric or by model choice, we decompose the variance in reweighting benefit ($\Delta\rho$) across 36 competent configurations via one-way ANOVA ($\eta^2$ for task vs model factors).

# 4. Experiments

## 4.1. Tasks and Datasets

We analyze four datasets representing 3 NLG evaluation tasks (*Summarization*, *Dialogue*, and *Essay*), with human annotations for both overall quality and traits (Table 1). To enable within-domain comparison for further investigation of rubric effects, we consider two essay datasets (ASAP++ and ELLIPSE). Details of trait definitions and annotation guidelines are in the Appendix.

**Annotations and Evaluation Rubric** All human annotations used in this study are external and originally reported with the datasets used. Each data set is published and peer-reviewed as a benchmark, with scores collected independently and prior to this work. We adopt every trait definition, scoring scale, and evaluation rubric as-is from the original dataset documentation (Stiennon et al., 2020; Mehri & Eskenazi, 2020; Mathias & Bhattacharyya, 2018; Crossley et al., 2023). No post-hoc rubric adjustment, tuning, or trait selection is performed during this study. In summary, the human annotations and evaluation rubric are externally validated benchmarks and our study introduces no new annotation protocol of its own. Detailed information about annotator recruitment, training, and inter-annotator agreement (where reported) can be found in the respective source publications. For ELLIPSE dataset, each essay was scored by two independent raters and the reported human score is their average. Inter-rater agreement is moderate across all six traits (Spearman $\rho$ from $0.62$ for vocabulary to $0.69$ for conventions), indicating that the human reference is not a single-annotator artifact.

## 4.2. LLM Judges

We evaluate four models (Claude Sonnet 4.5, GPT-5, Gemini 2.5 Pro, Mistral Large 2)[2], with 3 prompting strategies per task (rubric-based, attribute-first, chain-of-thought), totaling 36 configurations across three tasks. Additionally, we consider 3 configurations for ELLIPSE. For counterfactual experiments, we use LLaMA-3.1-8B and Claude Sonnet 4.5. Our experiments are organized into four groups corresponding to the research questions as follows.

**Cross-task experiment (RQ1, RQ2)** For each of the 36 configurations ($\sim$ 8100 evaluations), we compute the overall alignment ($\rho$, MAE, QWK). We primarily use Spearman's $\rho$ for our analysis. The QWK metric exhibits a similar pattern. trait-level alignment (per-trait $\rho$), decomposability (cross-validated $R^2$), attribution triangulation (Kendall's $\tau$ across three methods), and the ADT reweighting benefit ($\Delta\rho$ with bootstrap CI). The Variance decomposition ($\eta^2$) is calculated in the 36 configurations[3]. For robust estimation of trait-level alignment, compositionality, attribution divergence, we evaluate ELLIPSE essay dataset with three models (Claude, GPT, Gemini).

**Counterfactual perturbation (RQ3)** For each of the 5 ELLIPSE traits, we generate degraded versions of $\sim$ 300 essays using Claude with trait-specific instructions (e.g., for *grammar*: introduce subject-verb disagreements and tense inconsistencies while preserving ideas and structure). Both original and perturbed essays are scored by Claude and LLaMA on 5 traits plus overall quality. The magnitude of the perturbation varies by design: cohesion (57% character change) and conventions (43%) involve substantial rewrites; grammar (5%) and phraseology (9%) are more targeted; vocabulary (<0.1%, 95% of the essays unchanged) serves as a null control[4].

**Perturbation fidelity** We do not assume edits are perfectly isolated. We verify the perturbation localization as follows: 1) The perturbations are generated with trait-specific instructions, and their magnitude is measured per trait (character/word change and sequence similarity; refer to Appendix G), confirming the intended intensity and that the null control is near-zero; 2) It is observed that the non-target score

[2] We use Gemini for Gemini-2.5-pro, GPT for GPT-5, LLaMA for LLaMA-3.1-8B, and Claude for Claude-Sonnet-4.5 unless stated otherwise.

[3] USR's *understandable* trait is constant (all samples rated 1) in the persona subset and is excluded from the geometry; the rubric nominally has 5 traits. Geometry is computed on complete-case human scores ($n = 299/59/97/1031$).

[4] null control is a perturbation small enough to produce no detectable score change in a well-calibrated judge

| **Dataset** | **Task** | $N$ | $k$ | **Eff. Rank** | **Cond. #** | $R^2_{\text{human}}$ | **Max** $\rho_{\text{traits}}$ |
|---|---|---|---|---|---|---|---|
| TL;DR | Summarization | 350 | 3 | 2.78 | 2.31 | 0.885 | 0.32 |
| USR | Dialogue | 60 | 5 | 3.66 | 3.77 | 0.848 | 0.57 |
| ASAP++ | Essay | 100 | 5 | 1.74 | 47.3 | 0.547 | 0.91 |
| ELLIPSE | Essay | 1031 | 6 | 2.12 | 32.0 | 0.912 | 0.83 |

*Table 1.* Dataset and rubric characteristics. Effective rank and condition number are computed from the singular value decomposition of human trait score correlation matrices. A higher effective rank indicates more independent traits; a higher condition number indicates less stable weight estimation.

change is consistently smaller than the change for the target trait across both judges (refer to Non-target mean $\Delta$ in Table 5). This hints at the concentrated degradation on the intended dimension rather than the uniform spread; and 3) The vocabulary null control ($< 0.1\%$ character change) yields negligible drift, thus ruling out spurious global responses. The reported specificity (Target $\Delta$ minus Non-target $\Delta$) represents this localization, net of general drift.

**Cross-rubric control (RQ4)** Claude re-evaluates 100 ELLIPSE essays using the ASAP++ rubric (5 traits). We compare the resulting effective rank and condition number against: (a) ASAP++ on its own data, and (b) the same essays under the native ELLIPSE rubric.

**Evaluation pipeline** All judges use the default temperature. The prompts follow the published annotation guidelines, providing trait definitions and scoring scales. The output is a structured JSON with per-trait scores, overall score, and reasoning.

## 5. Results

We organize the analysis according to the proposed research questions. RQ1 establishes alignment at the overall and trait levels. RQ2 investigates whether overall judgments can be recovered from fine-grained trait scores. RQ3 examines whether these compositions are stable and human-aligned and uses human-weight recalibration (ADT) to diagnose aggregation mismatch. RQ4 studies when the rubric structure supports interpretable and diagnostically useful trait-level evaluation, using rubric geometry, cross-rubric control, and counterfactual validation. We report results following the four diagnostic stages of Figure 1 .

### RQ1: Alignment at overall and trait levels

We first evaluate the alignment of the LLM judge with the human in models, tasks and prompting strategies (refer to Figure 2). The strongest configuration varies by task, with no universally best prompting strategy (full per-task rankings in Appendix Table 4). This shows that multidimensional alignment is jointly shaped by task, rubric, and prompt format, rather than by a single, universally best prompting strategy.

Across 36 configurations (tasks x models x prompts), the mean overall-score alignment is Spearman $\rho$ = 0.542 [ 95% CI: 0.511, 0.573], while the mean trait-level alignment is 0.541 [95% CI: 0.512, 0.570]. However, fine-grained trait-level analysis uncovers a substantial divergence. Summarization achieves the strongest overall alignment of (0.577 - [ 95% CI: 0.507, 0.642]) but the weakest trait-level alignment (0.437 -[0.346, 0.522] ). In contrast, essay evaluations show higher trait-level alignment (0.619 - [0.484, 0.730]) but lower overall agreement (0.498 - [0.338, 0.630]). Dialogue is more balanced, with mean overall alignment of $\rho$ = 0.513 [0.297, 0.705] and mean trait-level alignment of $\rho$ = 0.534 [0.333, 0.698].

A further pattern is visible by attribute type: more verifiable traits such as coverage, accuracy, and uses_knowledge align more systematically with human judgments, while more subjective traits such as coherence and naturalness are comparatively weaker. Summarization is strongest on coverage ($\rho$ = 0.55) and weakest on coherence (0.24); dialogue is strongest on uses_knowledge (0.84) and weakest on naturalness (0.29). Essay traits are stronger and more uniform.

A deeper analysis of the Essay evaluation (ELLIPSE dataset) confirms the same pattern at higher resolution. On ELLIPSE, Gemini shows the strongest overall alignment with humans ($\rho$ = 0.739), followed by GPT (0.698) and Claude (0.681). At the trait level, mean correlation is 0.692 for Gemini, 0.637 for Claude, and 0.597 for GPT. Thus, even on a large rubric-rich essay dataset, overall agreement remains a weaker indicator of multidimensional alignment than trait-level evaluation.

### RQ2: Overall evaluations by LLM judges are strongly compositional across tasks

One of the primary objectives of this study is to investigate the interpretability of evaluation through decomposition. In the matched cross-task setting, linear trait-to-overall composition reports $R^2$ = 0.755 to 0.934, across all 36 configurations (Appendix Table 4) indicating that multidimensional LLM evaluation is usually decomposable, although the strength of this decomposition varies across rubric settings. These results suggest that overall judgments are not

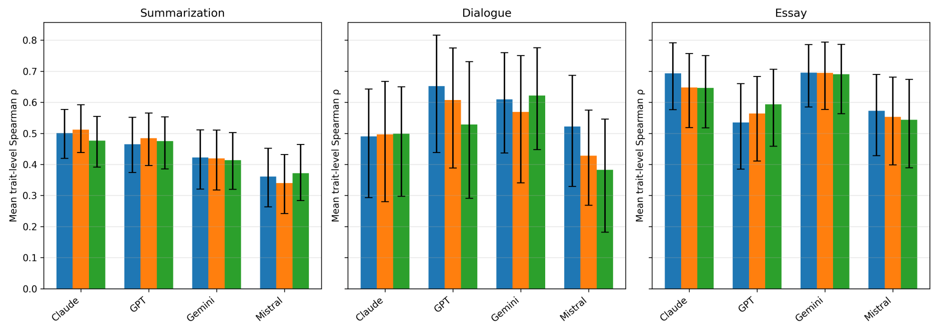


*Figure 2.* Mean alignment after bootstrapping across 3 tasks, 4 models, and 3 prompt strategies (Rubric: Blue, attribute_first: Orange, CoT: Green)

arbitrary outputs; they are largely systematic functions of fine-grained trait scores.

This pattern is stable across prompting strategies for all four judges (per-model averages in Appendix Table 4). Furthermore, this pattern is robust across attribution methods. Across tasks, linear/ridge coefficients, ablation, and SHAP recover similar broad compositional structures. In summarization, all three methods identify coverage as the dominant contributor for both humans and LLM judges. In dialogue and essay evaluation, the decomposition is more distributed, but the methods still converge on the broader structure of importance. This indicates that the observed trait–overall relationships are robust across complementary attribution methods.

The ELLIPSE analysis strengthens this conclusion in a controlled setting. Cross-validated ridge regression predicts the overall essay scores of the six traits with high accuracy for both humans and LLM judges: humans ($R^2$ = 0.912), and LLM judges (Gemini 0.916, Claude 0.915, GPT 0.766). This confirms that the ELLIPSE overall score is also structurally grounded in the trait profile rather than behaving as an independent output.

### RQ3: Decomposition is systematic but only partially human-aligned

Strong decomposability does not automatically guarantee human-aligned decomposition. In summarization, the human reference is strongly coverage-driven (trait attribution in composition as 81.0% coverage, 16.4% accuracy, 2.6% coherence). Rubric prompting comes closest to this pattern, especially for Gemini rubric, which differs only slightly from the human weighting (coverage -1.61, accuracy -0.90, coherence +2.51). In contrast, attribute-first and CoT assign greater importance to accuracy and coherence and less to coverage. For example, summarization-claude-attribute_first underweights coverage by -20.83 while overweighting accuracy (+7.25) and coherence (+13.58). A similar pattern is seen in both dialogue and essay evaluation. In dialogue, judges often overweight naturalness and engagingness while underweighting uses_knowledge and sometimes maintains_context. In essay evaluation, the attribution divergence is stronger. Across models and prompts, LLM judges repeatedly underweight content and organization, while overweighting word_choice, sentence_fluency, and conventions. Figure 4 shows the average attribution divergence across different tasks and models. These observations show that judges often learn stable but systematically different weighting or attribution rules from humans (Table 2(A)).

The detailed analysis of ELLIPSE confirms this more directly. The attribution methods agree with each other, with Kendall's $\tau$ ranging from 0.47 to 1.00 across ridge, SHAP, and ablation, but agreement between LLM and human trait ordering remains weak, with $\tau$ ranging from -0.07 to 0.20. Thus, judges are often internally consistent but externally misaligned. In other words, they decompose their own overall judgments in a stable way, but the decomposition is not similar to that of humans.

To diagnose whether this mismatch is caused by poor trait scoring or poor aggregation, we apply human-weight recalibration as an Alignment Diagnostic Test (ADT). Human-derived regression weights are applied to the LLM trait scores to form an adjusted overall score. Positive recalibration gains indicate compositional weight mismatch. In ELLIPSE, recalibration improves final-score alignment for all three judges, with gains of approximately +0.040 for Gemini, +0.076 for GPT, and +0.047 for Claude. This indicates that at least some part of the misalignment lies not in the trait scores themselves, but in how those scores are combined.

Causal sensitivity with counterfactual perturbations reveals a sharper alignment failure. Correlational composition ($R^2$, $\tau$) cannot distinguish a judge that genuinely responds to trait-level changes from one that produces compositionally-shaped outputs without underlying sensitivity. The counterfactual probe uncovers this distinction (Refer to Table

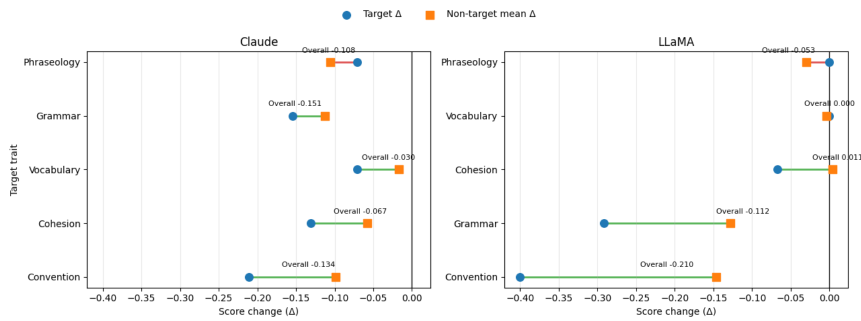


*Figure 3.* Counterfactual sensitivity heatmaps for Claude and LLaMA across five target traits. Rows correspond to targeted perturbations and columns report the change in the target trait score (Target Δ), the average change in non-target traits (Non-target mean Δ), the change in the overall score (Overall Δ), and specificity (Target Δ - Non-target mean Δ). More negative specificity indicates that the judge responds more strongly to the intended degraded trait than to the remaining traits, reflecting better targeted sensitivity.

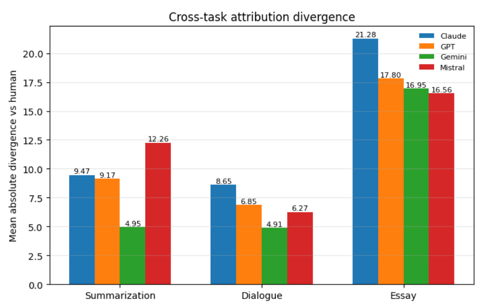


*Figure 4.* Cross-task attribution divergence from human composition. Bars show the mean absolute difference between LLM-inferred & human trait-importance patterns, averaged across traits and prompting strategies for each task–model pair. Higher values indicate larger departure from the human evaluative structure.

2(B), Figure 3). Claude responds specifically to targeted perturbations of conventions (specificity = -0.112) and cohesion (-0.072) but not to phraseology (+0.035, worse than the null baseline). In contrast, a far smaller LLaMA shows stronger specificity than Claude for conventions (-0.254) and grammar (-0.164), while both models fail in phraseology. Crucially, the null control (vocabulary) confirms that the scoring system is not responding spuriously: both models produce negligible shifts on a trait that was not meaningfully degraded, providing a calibration baseline for interpreting the sensitivity values of the other traits. This confirms that decomposable, correlationally-aligned outputs need not reflect compositional reasoning.

**RQ4: Rubric properties shape interpretability and diagnostic validity**

Next, we examine whether the observed differences in compositionality are tied to the structure of the evaluation rubric itself. Summarization is dominated by one dimension (coverage), dialogue is more distributed across several independent traits, and essay evaluation is more variable with most attributes being semi-objective.

Principal Component Analysis (PCA) on ELLIPSE trait scores shows that LLM judges produce more entangled trait spaces than humans. For Gemini and Claude, only 2 principal components from PCA explain 90% of the variance, compared with 3 components for humans and 4 for GPT. The effective rank is also lower for Gemini (1.77) and Claude (1.85) than the rank for humans (2.12) and GPT (3.24). This indicates substantial dependency among traits. Some rubric dimensions do not behave as independent signals. Under such conditions, a decomposition may look stable while still having a limited interpretive value.

The cross-rubric control shows that compositionality is tightly coupled with the evaluation rubric structure, not with the content. When ELLIPSE essay samples are evaluated using the ASAP++ rubric, the condition number remains essentially unchanged (49.8 to 50.2), matching ASAP++'s native condition number (47.3; Table 1) and showing that the instability is the result of rubric design and not the dataset. The Variance decomposition on the ADT signal shows that the task/rubric explains much more variance than the choice of the model ($\eta^2$ = 0.384 vs. 0.022).

## 6. Discussion

### 6.1. Rubric geometry, PCA, and trait dependencies

Rubric design plays an important role in content evaluation. It defines what traits are available and whether those traits behave like distinct evaluation dimensions for better interpretability. Our analysis shows that the trait spaces are essentially of lower dimensions. In particular, only a small number of principal components (2 to 4) is enough to explain most($\sim$ 90%) variance. Effective rank analyses indicate substantial dependency between traits. This means that some of the rubric-dimensions are not independent and work together as correlated proxies. A decomposition may look stable but have limited interpretability since multiple traits are partially substitutable. This dependency also explains

| Task | Claude | GPT | Gemini | Mistral |
|---|---|---|---|---|
| Summarization | 9.5 | 9.2 | 4.9 | 12.3 |
| Dialogue | 8.6 | 6.9 | 4.9 | 6.3 |
| Essay | 21.3 | 17.8 | 17.0 | 16.6 |

**(a) Attribution divergence**

| | Claude | | LLaMA | |
|---|---|---|---|---|
| Trait | Target Δ | Spec. | Target Δ | Spec. |
| Convention | -0.211 | **-0.112** | -0.400 | **-0.254** |
| Grammar | -0.154 | -0.041 | -0.292 | -0.164 |
| Cohesion | -0.131 | -0.073 | -0.067 | -0.071 |
| Vocabulary | -0.071 | -0.054 | 0.000 | 0.004 |
| Phraseology | -0.071 | 0.035 | 0.000 | 0.030 |

**(b) Counterfactual sensitivity**

*Table 2.* (a) Mean absolute divergence between LLM-inferred and human trait-importance patterns, averaged across traits and prompting strategies. Higher values indicate larger departure from human composition. (b) Counterfactual sensitivity summary. Target Δ is the change in the targeted trait score after degradation. Spec. (Specificity) is computed as Target Δ minus the mean non-target change; more negative values indicate better targeted detection. LLaMA is LLaMA-3.1-8B.

why richer rubric does not automatically enable better multidimensional evaluation. When traits are highly correlated, multidimensional outputs may overstate how much unique information each trait actually contributes. The Cross-rubric control experiment strengthens this interpretation. When ASAP++ rubric is applied to ELLIPSE essays, the structural instability largely persists, showing that geometry is more important than content while ensuring compositionality. This is the key reason rubric design should be treated as a primary factor, not a secondary annotation choice. In practice, before trusting trait-level outputs as independently meaningful, the rubric should be audited for effective rank, condition number, and the PCA dimensionality.

### 6.2. From diagnosis to practical mitigation

Our framework is diagnostic, but it provides concrete directions for mitigation. First, prompt strategy matters, i.e. rubric-based prompting is generally the safest option in summarization and essay evaluation, whereas dialogue is more model-dependent. Secondly, auditing the effective rank and condition number of the evaluation rubric from the human trait-correlation matrix provides additional information on possible mitigation strategies before deploying a multidimensional evaluation system. A high condition number together with an effective rank below the actual trait count $k$ indicates that the trait-level outputs are not independent signals and should not be trusted individually. Essay (ASAP++) is the clearest representative case where the condition number $47$ and the effective rank of $1.74$ over five traits (max pairwise $\rho = 0.91$, between *word_choice* and *sentence_fluency*) mean its nominal dimensions collapse towards roughly two. The mitigation strategy in such cases is to merge that collinear pair into a single "language use" dimension and further such pruning is necessary till the rubric approaches lower condition number. The third mitigation strategy is post-hoc recalibration when trait scores are already informative but are being aggregated with non-human weights. When trait scores are individually informative but their aggregation diverges from humans (denoted by a positive ADT gain, e.g. all three ELLIPSE judges ($+0.04$ to $+0.08\,\rho$)) , applying human-weight recalibration is appropriate rather than pruning. We observe that rubric-anchored prompting remains the safest option, while attribute-first and CoT prompting amplify aggregation mismatch.

### 6.3. Decomposability is not compositionality

A key observation of this study is that high decomposability and weak compositional alignment can coexist. The linear trait-to-overall score composition exhibits $R^2$ between 0.755 and 0.934, indicating LLM judges produce overall scores with stable decomposition into parts. Yet, the inferred composition diverges from the human compositional structure (Kendall's $\tau$ between -0.07 and 0.20), and counterfactual perturbations show that traits with strong correlation alignment with human evaluation often fail to respond when that trait is specifically degraded. The model's output appears compositional, but in reality, the compositional structure does not reflect the part-whole composition aligned with human experts. This follows the findings in compositional generalization research that compositionally-appearing outputs need not arise from a compositional mechanism (Hupkes et al., 2020; Dziri et al., 2023). Our results of the geometry of the rubric further highlight this by showing that an external compositional structure, such as the evaluation rubric, constrains the compositional behavior of the model 17x more than the capacity of the model ($\eta^2$ = 0.384 vs 0.022). This suggests that compositional behavior, at least for evaluation, is shaped more by the task than by the model's representational capacity.

## 7. Limitations

### Use of Linear Attribution Methods

Linear attribution methods may not fully capture the interplay among attributes. However, there is a distinction that needs to be considered. We use linear methods as a diagnostic tool, not as a predictive modeling tool, to infer how LLMs actually integrate fine-grained attribute information. We use linear methods as a theoretically optimal and interpretable choice for our diagnostic purposes. Theoretically, (Lundberg & Lee, 2017) prove that Shapley values-computed through linear approximations-provide the unique solution satisfying key properties for additive feature attribution: local accuracy, missingness, and consistency. Works such as (Parcalabescu & Frank, 2024; Horovicz & Goldshmidt, 2024) demonstrate SHAP acceptance and effectiveness in interpreting human-AI alignment for modern LLMs. Additionally, work in (Belinkov, 2022) establishes linear probes as the preferred standard in NLP interpretability research to avoid confounding probe capacity with model capabilities. Empirically, recent research (Chen et al., 2025) directly validates the linear approximation assumption. To validate the assumption of attribute discriminability required for linear decomposition, we examined pairwise correlations between ground-truth human ratings across all dimensions. Linearity is an informed choice for interpretability and not a cognitive claim about the model. Additionally, we investigated the nonlinear decomposition setting with SHAP (Random forest regressors) and Gradient Boosted Regression Trees (GBRT). We find that nonlinear compositional models do not significantly change the core assumption.

**Sample Size**

While our analysis setup spans across three diverse tasks (*Summarization*, *Dialogue*,*Essay*), the per-task samples sizes range from 58 to 350 placing inherent limits on statistical robustness. We have adequate power (1-$\beta \geq 0.80$) to detect large effects (Cohen's $d \geq 0.8$) but limited power for small effects ($d < 0.3$). Our bootstrapped confidence intervals account for uncertainty from limited samples, but future replication with N $>$ 200 per task would strengthen statistical conclusions. We use larger ELLIPSE essay dataset for detailed analysis.

**Prompt Coverage**

We test three distinct prompting strategies (rubric-based, attribute-first, CoT), each strategy with fixed phrasing. Our task-level findings are stable across these three strategies, but we cannot claim robustness to arbitrary prompt variations. Future work should systematically vary prompt phrasing within each strategy type to assess sensitivity.

**Task Coverage**

We consider three tasks for our initial study, but broader validation across additional NLG types would strengthen generalization. Tasks such as machine translation, data-to-text generation, and code generation exhibit unique multicollinearity patterns requiring out-of-the-box mitigation strategies.

**Annotator Coverage**

As noted in §4.1, our human annotations are taken directly from the published benchmarks. Accordingly, per-sample annotation counts and agreement follow each source's protocol. We agree that multi-annotator data throughout would nevertheless add confidence.

**Behavioral Analysis**

Our study focuses on input trait scores and overall evaluations. We characterize how judges combine fine-grained scores into a holistic score. We do not consider mechanistic analysis at the token-level or internal model representations. It is a complementary direction that could explain *why* certain traits are harder to judge or perturb.

## 8. Conclusion

LLM judges have emerged as a scalable alternative to human evaluation, but reliable multidimensional alignment requires more than high overall agreement. Across three NLG tasks, we show that overall scores are strongly recoverable from trait profiles ($R^2$ = 0.755 to 0.934), yet this decomposability does not imply human-aligned compositional reasoning. LLM judges are often internally consistent while systematically diverging from human evaluative structure ($\tau$ = -0.07 to 0.20) and counterfactual perturbations confirm that traits with strong correlational alignment can fail to respond when specifically degraded — decomposable outputs are not the same as compositionally-grounded ones. Our Alignment Diagnostic Test reveals that a substantial component of this misalignment is compositional — correctable through recalibration — rather than arising purely from poor trait scoring. The primary determinant of whether multidimensional evaluation delivers genuine diagnostic value is rubric geometry. Rubrics with higher effective rank and lower condition numbers support stable, interpretable, and human-aligned compositions and entangled rubrics produce misleadingly structured outputs regardless of model capability. Counterfactual perturbation experiments confirm that correlational alignment overestimates causal sensitivity, particularly for subjective traits with low trait independence. The external compositional structure imposed by the rubric constrains compositional behavior 17× more than model choice ($\eta^2$= 0.384 vs. 0.022) for evaluation and the task's compositional structure dominates the model's representational capacity. These findings suggest a practical workflow: before deploying multidimensional LLM evaluation, audit rubric effective rank and condition number; use rubric-anchored prompting; and apply ADT to distinguish scoring from aggregation failures. Trustworthy evaluation requires not only that judges agree with humans, but that they do so for the right compositional reasons.

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# A. Detailed Related Work

## A.1. LLM-as-a-Judge

LLMs as evaluators- now referred to as LLM-as-a-Judge-are being used to determine whether output is within an established scope. Various surveys (Gu et al., 2024; Liu et al., 2023) provide systematic insights into dimensions such as formal definition and formulation, applications, and limitations. Recent works have demonstrated the use of LLM judges in diverse applications such as story evaluation (Chiang & Lee, 2023), code comprehension(Yuan et al., 2023) and general open-ended tasks like Chatbot Arena and MT-Bench(Zheng et al., 2023). Despite their promising performance and high alignment with human evaluators in diverse settings (Kim et al., 2024; Liu et al., 2023; Zheng et al., 2023; Sottana et al., 2023), even powerful LLM judges still exhibit known issues of hallucination (Ye et al., 2023; Huang et al., 2025), factual errors (Wang et al., 2024; Turpin et al., 2023), and bias(Gallegos et al., 2024). Almost all these works report aggregate evaluation (overall $\rho$, accuracy, $K$). We extend this work beyond the overall score by asking how do judges compose their holistic judgments from fine-grained criteria?

## A.2. Multidimensional Evaluation

Previous evaluation studies focused primarily on coarse-grained evaluations, limiting interpretability. Recent studies like G-Eval have studied multidimensional evaluation in their framework. G-Eval(Liu et al., 2023) proposes a foundational framework that uses GPT-4-turbo with chain-of-thought reasoning to decompose given evaluation criteria into structured steps. (Feng et al., 2025) introduces SaMer, a scenario-aware multidimensional evaluator that is capable of adapting evaluation dimensions based on query context. Works such as FLASK (Fine-Grained Language Model Evaluation)(Ye et al., 2024), LMUnit(Saad-Falcon et al., 2024), HD-EVAL (Liu et al., 2024), and (Yu et al., 2025) demonstrate the effectiveness of decomposed evaluation enabling interpretable assessment and improved human-LLM evaluation agreement. However, these frameworks focus on improving evaluation quality through structured criteria, not on diagnosing why and where misalignment occurs. These frameworks use decomposition to improve evaluation quality by designing better rubrics or better prompting. None of them systematically explore the compositional relationship between traits and overall scores or explore if decomposition adds genuine diagnostic value. We complement these approaches by providing analytical tools to determine whether decomposition is worthwhile for a given rubric before deploying it.

## A.3. Attribution and Interpretability

Recent work on LLM explainability has focused on different granularities of interpretation. SHAP (Lundberg & Lee, 2017) and LIME (Ribeiro et al., 2016) provide foundational feature attribution methods, token-level attribution methods (Horovicz & Goldshmidt, 2024) apply SHAP to individual tokens, while (Chen et al., 2025) extract locally linear decision surfaces from LLM internal representations. Attribution-informed methods have been successfully explored for human-AI alignment. (Turner et al., 2024) introduces activation engineering that modifies activations in inference time to steer the model outputs in a specific direction. They suggested that activation engineering can flexibly re-target LLM behavior without damaging general performance. The SHAP / LIME-based dense reward shaping framework in (Koo et al., 2025) shows a 10–15% improvement in the RLHF pipeline for LLM alignment. This technique directly addresses credit assignment, precisely the challenge facing multidimensional evaluation agents that must attribute overall quality to specific attributes such as helpfulness, harmlessness, and honesty. Previous work considers attribution at token or representation level. We apply attribution methods at the evaluation trait level.

# B. Formulation

*Table 3.* Formulation

| | | | |
|---|---|---|---|
| $y_{overall^l}$ | overall score by an LLM judge | $y_{overall^h} \in [1,5]$ | overall score by human |
| $a_{j^h}(x_i)$ | human score for attribute $j$ | $a_{j^l}(x_i)$ | LLM judge score for attribute $j$ |
| $A = \{a_1, .., a_m\}$ | $m$ attributes for evaluation | | |
| $\rho(y^h, y^l)$ | Spearman's correlation between scores | $w^h(a_j) \neq w^l(a_j)$ | LLM judge weighs attribute contributions differently |
| $a_{j^h} \neq a_{j^l}$ | attribute-score divergence | | |

## C. Tasks, Datasets, and Rubrics

We analyze four datasets spanning three NLG domains. Each dataset provides human annotations for both overall quality and fine-grained traits, and is drawn from a published, peer-reviewed benchmark.

**Summarization (TL;DR).** 350 samples from the TL;DR dataset (Stiennon et al., 2020), evaluated on {*coverage, accuracy, coherence*}. The source document provides a grounding reference for verification. Evaluation prompts follow the annotation guidelines reported in (Stiennon et al., 2020).

**Dialogue (USR).** 58–60 persona-based dialogue samples from the USR dataset (Mehri & Eskenazi, 2020), evaluated on {*understandable, natural, maintains_context, engaging, uses_knowledge*}. This task offers minimal grounding—most traits depend on subjective and contextual judgment. We use the complete 'persona' subset.

**Essay – ASAP++.** 97–100 essays from the ASAP++ dataset (Mathias & Bhattacharyya, 2018), evaluated on {*content, organization, word_choice, sentence_fluency, conventions*}. This rubric produces highly intercorrelated traits (max pairwise $\rho = 0.91$), making it a natural test case for rubric-driven compositional effects.

**Essay – ELLIPSE.** 1,031 essays from the ELLIPSE corpus (Crossley et al., 2023), evaluated on {*cohesion, syntax, vocabulary, phraseology, grammar, conventions*}. This dataset is substantially larger than the others ($3\times$ the next largest) and uses a rubric with more independent traits (effective rank 2.12 vs. ASAP++'s 1.74), enabling robust statistical estimates and a natural contrast within the same task domain (essay evaluation).

## D. Attribute Types

- Verifiable: These traits have concrete, checkable referents in the source material: accuracy (can verify claims against source), coverage (can check which key points are included), uses_knowledge (can verify factual content), content (can assess idea coverage against prompt). These consistently report higher alignment ($\rho$ = 0.55–0.82).

- Semi-Objective: Semi-objective traits are intermediate — they have partial verifiable markers but also require judgment: conventions (grammar/spelling errors are checkable, but what counts as "good conventions" is partly subjective), organization (structure can be observed, but quality of structure is judged).

- Subjective:Subjective traits generally represent holistic judgment without a concrete external referent: coherence (logical flow — no checklist), natural (human-likeness — inherently subjective), engaging (reader experience), phraseology (expression quality). These consistently achieve lower alignment ($\rho$ = 0.29–0.48).

## E. LLM Judges

For each model-task combination, we test three prompting strategies:

**Rubric-based** We provide a detailed definition and explicit scoring criteria for all fine-grained evaluation measures and the overall quality measure.

**Attribute-First** In this setup, we explicitly instruct LLM judges to score fine-grained measures before evaluating the overall quality, forcing bottom-up aggregation.

**Chain-Of-Thought (CoT)** We introduce explicit step-by-step reasoning for every dimension before scoring.

We follow the annotation guidelines provided in the relevant work to design evaluation prompts for LLM judges. We adopt evaluation frameworks from dataset documentation (Stiennon et al., 2020; Mehri & Eskenazi, 2020; Mathias & Bhattacharyya, 2018), which are used and validated through previous works. Output is returned as structured JSON containing scores and reasoning for each trait and overall quality.

We define *competent* models as those achieving $\rho \geq 0.3$ with human overall scores across configurations, following the rationale that attribution analysis of non-systematic evaluators would not reveal meaningful compositional patterns.

## F. Rubric Geometry

We characterize rubrics through two structural properties derived from the singular value decomposition (SVD) of the trait score correlation matrix:

**Effective rank.** The number of independent dimensions in the trait space, computed as the exponential of the Shannon entropy of the normalized singular values:

$$\text{eff\_rank} = \exp\left(-\sum_i p_i \log p_i\right), \quad p_i = \sigma_i / \textstyle\sum_j \sigma_j \tag{1}$$

An effective rank of 1.0 means all traits measure essentially the same underlying quality; an effective rank equal to $k$ (the number of traits) means all traits are fully independent. This provides a continuous, principled measure of trait independence.

**Condition number.** The ratio of the largest to the smallest singular value ($\sigma_{\max}/\sigma_{\min}$), indicating the numerical stability of weight estimation. When the condition number is high, multiple weight configurations fit the data equally well, making it impossible to reliably determine which traits the evaluator considers most important..

## G. Counterfactual Validation

Correlational alignment establishes *association* between LLM and human trait scores but not *causal sensitivity*: an LLM judge might assign "cohesion" scores based on surface cues that correlate with cohesion in the training distribution without causally responding to cohesion-specific quality changes. To test causal sensitivity, we construct counterfactual perturbation experiments on the ELLIPSE dataset.

**Perturbation generation.** For each of 5 target traits (cohesion, conventions, grammar, phraseology, vocabulary) and each of 300 essays, we generate a perturbed version that specifically degrades the target trait while preserving other qualities. Perturbations are generated by rules with trait-specific degradation instructions. We measure perturbation magnitude through character change percentage, word change percentage, and sequence similarity ratio, confirming that perturbations vary in intensity across traits (from $<$1% for vocabulary to 57% for cohesion).

**Evaluation.** Both original and perturbed essays are scored by two LLM judges: Claude Sonnet (a competent judge from our main analysis) and LLaMA-3.1-8B (a smaller model, testing the effect of capacity). Judges score 5 traits plus overall quality for each essay.

*Table 4.* Trait-to-overall compositionality across tasks, models, and prompts

| Task | Model | Prompt | $\rho$ | MAE | Within-1 | Bias | $R^2$ | Top Dimension |
|---|---|---|---|---|---|---|---|---|
| Dialogue | Claude | CoT | 0.587 | 1.638 | 40.00% | 1.638 | 0.834 | engaging (34.8%) |
| Dialogue | Claude | Rubric | 0.496 | 1.383 | 58.30% | 1.383 | 0.764 | natural (28.8%) |
| Dialogue | Claude | attribute-first | 0.572 | 1.475 | 51.70% | 1.475 | 0.797 | engaging (36.8%) |
| Dialogue | GPT5 | CoT | 0.5 | 1.017 | 78.30% | 0.983 | 0.857 | maintains_context (31.3%) |
| Dialogue | GPT5 | Rubric | 0.571 | 0.917 | 88.30% | 0.85 | 0.676 | maintains_context (32.0%) |
| Dialogue | GPT5 | attribute-first | 0.626 | 0.867 | 86.70% | 0.833 | 0.922 | engaging (36.4%) |
| Dialogue | Gemini | CoT | 0.515 | 1.1 | 71.70% | 0.933 | 0.844 | engaging (30.9%) |
| Dialogue | Gemini | Rubric | 0.633 | 0.767 | 88.30% | 0.733 | 0.888 | engaging (34.0%) |
| Dialogue | Gemini | attribute-first | 0.418 | 1.15 | 73.30% | 0.95 | 0.883 | engaging (33.6%) |
| Dialogue | Mistral | CoT | 0.418 | 1.593 | 44.10% | 1.593 | 0.687 | uses_knowledge (29.6%) |
| Dialogue | Mistral | Rubric | 0.377 | 1.136 | 71.20% | 1.102 | 0.717 | natural (34.2%) |
| Dialogue | Mistral | attribute-first | 0.448 | 1.322 | 62.70% | 1.322 | 0.720 | engaging (41.6%) |
| Essay | Claude | CoT | 0.522 | 1.299 | 60.80% | 1.299 | 0.939 | sentence_fluency (68.4%) |
| Essay | Claude | Rubric | 0.577 | 0.845 | 85.60% | 0.845 | 0.934 | sentence_fluency (38.5%) |
| Essay | Claude | attribute-first | 0.534 | 1.237 | 67.00% | 1.237 | 0.856 | sentence_fluency (51.4%) |
| Essay | GPT5 | CoT | 0.531 | 1.485 | 46.40% | 1.485 | 0.675 | word_choice (43.3%) |
| Essay | GPT5 | Rubric | 0.346 | 1.01 | 76.30% | 0.969 | 0.784 | word_choice (57.0%) |
| Essay | GPT5 | attribute-first | 0.566 | 1.495 | 45.40% | 1.495 | 0.805 | word_choice (50.4%) |
| Essay | Gemini | CoT | 0.508 | 1.32 | 58.80% | 1.32 | 0.882 | word_choice (50.6%) |
| Essay | Gemini | Rubric | 0.628 | 0.784 | 89.70% | 0.784 | 0.842 | sentence_fluency (24.9%) |
| Essay | Gemini | attribute-first | 0.565 | 1.361 | 58.80% | 1.361 | 0.884 | sentence_fluency (40.0%) |
| Essay | Mistral | CoT | 0.385 | 1.093 | 73.20% | 1.052 | 0.806 | organization (31.6%) |
| Essay | Mistral | Rubric | 0.45 | 0.701 | 92.80% | 0.515 | 0.805 | sentence_fluency (25.4%) |
| Essay | Mistral | attribute-first | 0.431 | 0.938 | 78.40% | 0.938 | 0.812 | sentence_fluency (44.7%) |
| SUMM | Claude | CoT | 0.686 | 1.185 | 65.80% | 0.639 | 0.931 | coverage (64.9%) |
| SUMM | Claude | Rubric | 0.704 | 1.198 | 62.90% | 0.611 | 0.945 | coverage (82.7%) |
| SUMM | Claude | attribute-first | 0.679 | 1.144 | 67.20% | 0.357 | 0.909 | coverage (60.2%) |
| SUMM | GPT5 | CoT | 0.621 | 1.234 | 63.50% | 0.273 | 0.909 | coverage (66.1%) |
| SUMM | GPT5 | Rubric | 0.65 | 1.229 | 62.90% | 0.38 | 0.869 | coverage (87.4%) |
| SUMM | GPT5 | attribute-first | 0.632 | 1.215 | 64.90% | 0.012 | 0.900 | coverage (63.2%) |
| SUMM | Gemini | CoT | 0.615 | 1.426 | 58.20% | 0.623 | 0.919 | coverage (72.4%) |
| SUMM | Gemini | Rubric | 0.695 | 1.212 | 65.20% | 0.572 | 0.891 | coverage (79.4%) |
| SUMM | Gemini | attribute-first | 0.614 | 1.36 | 59.90% | 0.47 | 0.907 | coverage (69.9%) |
| SUMM | Mistral | CoT | 0.577 | 1.344 | 60.90% | -0.301 | 0.876 | coverage (70.6%) |
| SUMM | Mistral | Rubric | 0.473 | 1.495 | 51.50% | -0.09 | 0.822 | coverage (90.1%) |
| SUMM | Mistral | attribute-first | 0.473 | 1.629 | 53.50% | -1.107 | 0.844 | coverage (49.9%) |

*Table 5.* Counterfactual diagnostic validity results. Negative specificity indicates that the targeted trait drops more than the average non-target traits.

| Judge | Target trait | Target Δ [95% CI] | Non-target mean Δ | Specificity [95% CI] | Overall Δ |
|---|---|---|---|---|---|
| LLaMA-3.1-8B | conventions | -0.400 [-0.570, -0.240] | -0.146 | -0.254 [-0.370, -0.136] | -0.21 |
| LLaMA-3.1-8B | grammar | -0.292 [-0.416, -0.180] | -0.128 | -0.164 [-0.252, -0.081] | -0.11236 |
| LLaMA-3.1-8B | cohesion | -0.067 [-0.157, 0.011] | 0.004 | -0.072 [-0.155, 0.007] | 0.011236 |
| LLaMA-3.1-8B | vocabulary | 0.000 [0.000, 0.000] | -0.004 | 0.004 [0.000, 0.013] | 0 |
| LLaMA-3.1-8B | phraseology | 0.000 [-0.106, 0.096] | -0.030 | 0.030 [-0.036, 0.096] | -0.05319 |
| Claude | cohesion | -0.131 [-0.188, -0.074] | -0.058 | -0.072 [-0.121, -0.023] | -0.06711 |
| Claude | conventions | -0.211 [-0.268, -0.154] | -0.099 | -0.112 [-0.160, -0.064] | -0.13378 |
| Claude | grammar | -0.154 [-0.204, -0.104] | -0.113 | -0.041 [-0.084, 0.003] | -0.1505 |
| Claude | phraseology | -0.071 [-0.125, -0.017] | -0.106 | 0.035 [-0.013, 0.082] | -0.10774 |
| Claude | vocabulary | -0.071 [-0.125, -0.017] | -0.017 | -0.054 [-0.096, -0.013] | -0.0303 |

*Table 6.* Attribution divergence with respect to human across models and tasks

| Task | Model | Prompt | MeanAbsGap | MaxAbsGap |
|---|---|---|---|---|
| Dialogue | Claude | Attribute-first | 9.58617 | 11.17741 |
| Dialogue | Claude | CoT | 10.51147 | 14.54501 |
| Dialogue | Claude | Rubric | 5.845067 | 11.69013 |
| Dialogue | GPT | Attribute-first | 6.582695 | 12.61899 |
| Dialogue | GPT | CoT | 8.008508 | 16.01702 |
| Dialogue | GPT | Rubric | 5.970185 | 6.587025 |
| Dialogue | Gemini | Attribute-first | 4.472784 | 7.105544 |
| Dialogue | Gemini | CoT | 3.810048 | 7.620096 |
| Dialogue | Gemini | Rubric | 6.435775 | 11.58014 |
| Dialogue | Mistral | Attribute-first | 7.546183 | 15.09237 |
| Dialogue | Mistral | CoT | 2.7277 | 5.450091 |
| Dialogue | Mistral | Rubric | 8.524175 | 17.04835 |
| Essay | Claude | Attribute-first | 19.58394 | 41.73973 |
| Essay | Claude | CoT | 24.15037 | 58.76696 |
| Essay | Claude | Rubric | 20.10106 | 41.45862 |
| Essay | GPT | Attribute-first | 16.90706 | 41.36806 |
| Essay | GPT | CoT | 14.18025 | 34.26387 |
| Essay | GPT | Rubric | 22.31621 | 47.95149 |
| Essay | Gemini | Attribute-first | 17.82395 | 30.37357 |
| Essay | Gemini | CoT | 18.97092 | 41.55763 |
| Essay | Gemini | Rubric | 14.05708 | 28.03101 |
| Essay | Mistral | Attribute-first | 19.96702 | 35.22933 |
| Essay | Mistral | CoT | 14.83902 | 36.74466 |
| Essay | Mistral | Rubric | 14.86442 | 32.36997 |
| Summarization | Claude | Attribute-first | 13.88609 | 20.82913 |
| Summarization | Claude | CoT | 10.79387 | 16.1908 |
| Summarization | Claude | Rubric | 3.736321 | 5.604481 |
| Summarization | GPT | Attribute-first | 11.91422 | 17.87133 |
| Summarization | GPT | CoT | 9.973803 | 14.9607 |
| Summarization | GPT | Rubric | 5.611594 | 8.417392 |
| Summarization | Gemini | Attribute-first | 7.400979 | 11.10147 |
| Summarization | Gemini | CoT | 5.770958 | 8.656437 |
| Summarization | Gemini | Rubric | 1.674108 | 2.511162 |
| Summarization | Mistral | Attribute-first | 20.79654 | 31.19481 |
| Summarization | Mistral | CoT | 9.335588 | 14.00338 |
| Summarization | Mistral | Rubric | 6.656842 | 9.985262 |